

Taeho Kim

Physics Graduate Student

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EDUCATION

Princeton University, 2022-2026

B.A. Astrophysics with Honors, Mathematics Minor

Gravitational Waves Data Group Member

Relevant Coursework: Condensed Matter I & II, Dynamics of Stellar & Planetary Systems, Tensor Calculus, Differential Geometry, Algebraic Geometry, Algebraic Topology, Real & Complex Analysis, Quantum I & II, General Relativity, Cosmology, Gravitational Waves, Thermodynamics & Statistical Mechanics

Washington University in St. Louis, 2026-2031

Ph.D. Physics, Hughes Fellow

Advisors: Prof. Alexander Seidel

Relevant Coursework: Quantum Field Theory I (RC), Functional Analysis & Measure Theory I, Supervised Research (Prof. Alexander Seidel), Electrodynamics

RESEARCH INTERESTS

Mathematical Physics, Galactic Center Populations, Topological States of Matter, High Energy Physics, Gravitational Waves and Neutron Star EoS.

PUBLICATIONS

On the Missing Red Giants at the Galactic Center

Taeho Kim, Jeremy Goodman. 2026. The Astrophysical Journal, Vol. 1001, 109. ArXiv [3 citations]

Spatial and spectral constraints on resolved mass-loss of the massive Post-RSG star IRAS 17163–3907 and its Fried Egg Nebula

E. Koumpia, A. Cikota, W.-J. de Wit, G. Muñoz-Sanchez, **T. Kim**, A. Corporaal, R. D. Oudmaijer, *et al* (8 other co-authors). 2026. Astronomy & Astrophysics, Vol. 709, A83. ArXiv [1 citation]

Dynamical Tides and Gravitational Waveforms of Binary Neutron Star Mergers

Taeho Kim, Abhishek Hegade, Sylvia Biscoveanu. 2026. [Undergraduate Thesis]

Galactic Center Red Giants

Jeremy Goodman, **Taeho Kim**. 2027. [in preparation]

Dissecting the Morphology of the Multiphase Galactic Web

Taeho Kim, Chang-goo Kim, James Sunseri. 2027. [in preparation]

On the Hopf Fibration and Hopfions

Taeho Kim. 2027. [expository, in preparation]

POSTERS & TALKS

The First L,M Band Analysis of Post-RSG Yellow Hypergiants

ALMA, ESO Summer Research Presentations. Contributed Talk (AUG 2023)

ESO Hypergiant Team Research Presentation. Contributed Talk (OCT 2023)

Dissecting the Morphology of the Multiphase Galactic Web

Astrophysics Summer Research Symposium. Contributed Talk (AUG 2025)

An Examination of Missing Red Giants near the Galactic Center

American Astronomical Society 247 Meeting. Contributed iPoster (JAN 2026)

AWARDS & GRANTS

Arthur L. Hughes Fellowship

Washington University in St. Louis (AUG 2026)

Class of 1955 Senior Thesis Fund Recipient

Princeton University (JAN 2026)

Chilean Mathematical Olympiad Silver Medalist

Sociedad de Matemática de Chile, Invitation to Chilean IMO Team Tryouts, 12 Invited Nation-wide (DEC 2021)

TEACHING

Graduate Theoretical Physics II (PHYS5020)

Assistant Instructor

JAN 2027 - MAY 2027 (scheduled), Washington University in St. Louis, Department of Physics

Graduate Theoretical Physics I (PHYS5010)

Part-time Assistant Instructor

SEP 2026 - DEC 2026, Washington University in St. Louis, Department of Physics

General Relativity (AST301/PHY321)

Undergraduate Course Assistant & Grader

SEP 2025 - DEC 2025, Princeton University, Department of Astrophysical Sciences

Real Analysis [Lebesgue Measure and Integration] (MAT320)

Undergraduate Course Assistant

SEP 2025 - DEC 2025, Princeton University, Department of Mathematics

Thermodynamics and Statistical Mechanics (PHY301)

Physics Tutor, Undergraduate Physics Tutoring

SEP 2025 - DEC 2025, Princeton University, Department of Physics

OTHER EXPERIENCE

SPLASH Princeton, Coordinator

FEB 2023 - APR 2025, Princeton University

Coordinated Princeton SPLASH 2024 (funding, outreach, materials)

Instructor for Princeton SPLASH 2023, 2024

Nelson Lab, Undergraduate Research Assistant

JUN 2024 - JUN 2025, Princeton University, Department of Chemical & Biological Engineering

Firestone Borrow Direct Employee

JUN 2024 - DEC 2025, Princeton University, Firestone Library